Twice the stone.

How much more edge?

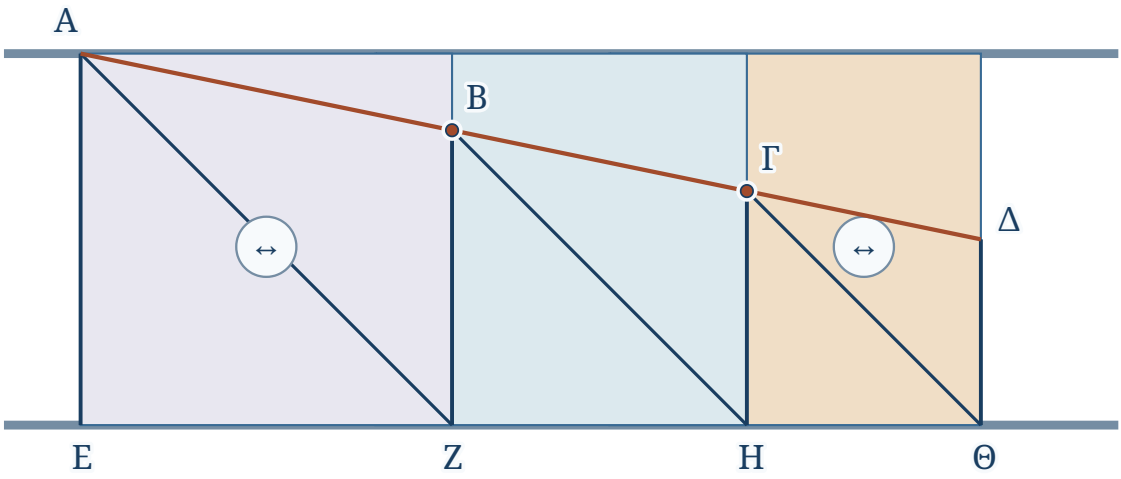
Saved ideal pose. Required volume 2 × the original; edge ΔΘ = 1 unit. Equal square plates and the numerical dimensions are modern teaching choices.

Open this pose in the workshop · Save these measurements · Save the A4 paper sheet

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Your selected machine

Left slide 20.6299474%; right slide 37.0039475%. The four points are collinear in this ideal mathematical model.

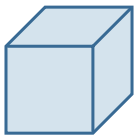


Left plate above middle · right plate below middle · equal square panels

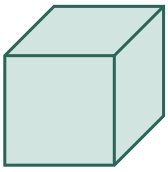
In a chosen 60 mm frame, the signed vertical offsets from the line AΔ are B 0 mm and Γ 0 mm. They run along the panel edges in the horizontal orientation.

Four lengths in units of the original cube edge; read the last mean ΓH for the new edge.

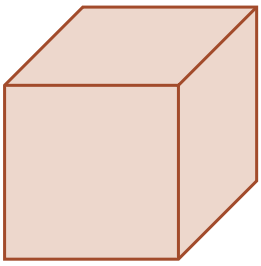
Length	Given / contact	Value	Ratio to next
a · AE	Given	2	1.259921
b · BZ	Exposed contact	1.5874	1.259921
c · ΓH	Exposed contact	1.25992	1.259921
d · ΔΘ	Given	1	—



Original
edge 1



Your ΓH
edge 1.25992

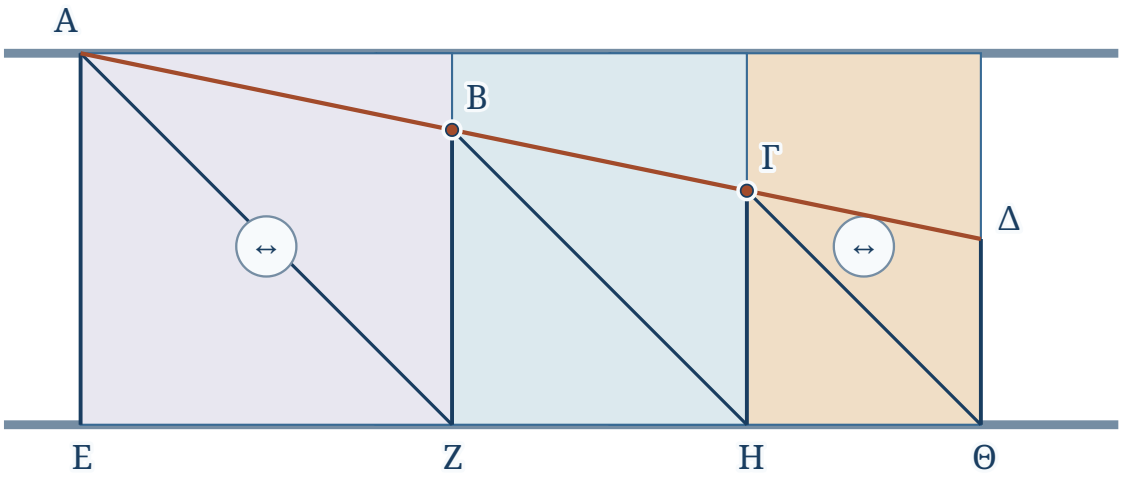


Double edge
edge 2

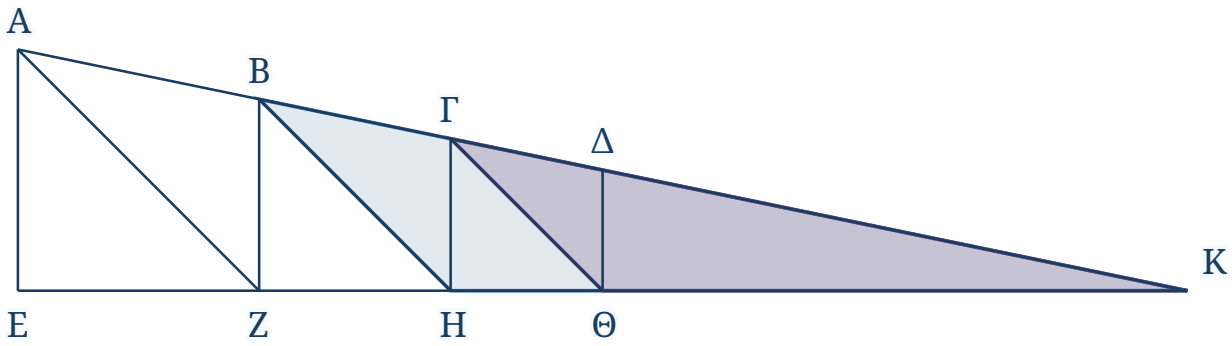
The candidate edge ΓH is 1.25992. Its cube has volume 2 × the original, with signed error 0% relative to the requested 2 ×. Doubling every edge would give eightfold volume.

Compare the ideal alignment and its proof

This separate comparison calculates both contacts from the cube root. It does not replace your selected pose above. Write $t = \sqrt[3]{(1/R)}$. The panel heights become 1, t, t^2, t^3; multiplying them by R makes ΔΘ equal to one original cube edge.



Left plate above middle · right plate below middle · equal square panels



1. Four lengths

AE and ΔΘ are given. The exposed contacts B and Γ supply candidate lengths BZ and ΓH. Move both plates until all four upper points lie on the straightedge.

2. Vertical parallels

With A, B, Γ, Δ on the line to K, vertical parallels give AK/BK = EK/ZK = AE/BZ. The blue triangles share their angle at K.

3. Diagonal parallels

The parallel diagonals AZ and BH also give AK/BK = ZK/HK. So EK/ZK = ZK/HK. The purple triangles use the same sloping upper line, with different lower sides.

4. Repeat at Γ

Repeat with B, Γ: vertical parallels give BK/ΓK = ZK/HK; parallel diagonals BH and ΓΘ give BK/ΓK = HK/ΘK. Thus all three successive baseline ratios agree, and so do the corresponding vertical ratios.

5. A cube follows

Write the common ratio as r. Then a/d = (a/b)(b/c)(c/d) = r^3. The new edge is c/d = r times the old edge; its cube has a/d times the volume. The source's ratios become the modern algebra shown here.

Thus for R=2, the required edge is 1.25992. All measurements use ordinary floating-point arithmetic; the diagram and agreement do not replace the geometric proof.

Read eight selected source passages

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1. Double which magnitude?

τῶν γὰρ πλευρῶν διπλασιασθεισῶν τὸ μὲν ἐπίπεδον γίνεται τετραπλάσιον, τὸ δὲ στερεὸν ὀκταπλάσιον.

nam lateribus conduplicatis planum quadruplo maius fit, solidum uero octuplo maius.

For when the sides have been doubled, the plane figure becomes fourfold, but the solid eightfold.

διπλασιασθεισῶν · conduplicatis

The Greek genitive absolute makes the doubled sides a condition for what follows; Latin uses an ablative absolute. τὸ μὲν … τὸ δὲ contrasts the plane figure with the solid. Eutocius transmits this correction within a story about Minos and a tomb. The story is a literary report, not evidence of an excavated building.

Printed pp.104, 105.

2. Two means, one continuing ratio

δεδόσθωσαν δύο ἄνισοι εὐθεῖαι, ὧν δεῖ δύο μέσας ἀνάλογον εὑρεῖν ἐν συνεχεῖ ἀναλογίᾳ, αἱ ΑΕ, ΔΘ.

datae sint duae lineae inaequales, inter quas duas medias proportionales inueniri oportet in proportione continua, AE, ΔΘ.

Let two unequal straight lines, AE and ΔΘ, be given, between which two mean proportionals must be found in a continuing proportion.

μέσας ἀνάλογον · medias proportionales

μέσας is feminine plural: supply εὐθείας, straight lines. A mean here is an intermediate length, rather than the arithmetic average. ἐν συνεχεῖ ἀναλογίᾳ requires one continuing ratio: a:b = b:c = c:d. The next command places AE perpendicular to the baseline; this is where the p.109 diagram begins.

Printed pp.106, 107.

3. Keep the middle fixed

μένοντος δὴ τοῦ μέσου παραλληλογράμμου τοῦ ZI συνωσθήτω τὸ μὲν AZ ἐπάνω τοῦ μέσου, τὸ δὲ IΘ ὑποκάτω,

itaque manente parallelogrammo medio ZI inter se adpropinquent AZ supra medium, IΘ infra,

While the middle parallelogram ZI remains in place, let AZ be pushed together above the middle one, and IΘ below it.

μένοντος · manente

Another absolute construction gives the condition during the movement. ἐπάνω / ὑποκάτω specifies the layers of the plates. The diagonal of the next plate becomes visible at each preceding plate's edge. The parallelograms are named by opposite corners; AZ is also a diagonal. This workshop chooses square panels.

Printed pp.108, 109.

4. Until all four are on one line

καθάπερ ἐπὶ τοῦ δευτέρου σχήματος, ἕως οὗ γένηται τὰ Α, Β, Γ, Δ κατ᾽ εὐθεῖαν, καὶ διήχθω διὰ τῶν Α, Β, Γ, Δ σημείων εὐθεῖα καὶ συμπιπτέτω τῇ ΕΘ ἐκβληθείσῃ κατὰ τὸ Κ.

ut in figura secunda, dum puncta A, B, Γ, Δ in eadem linea recta sint, et per A, B, Γ, Δ puncta ducatur linea et concurrat cum EΘ producta in K.

As in the second figure, until A, B, Γ and Δ lie on a straight line; draw a straight line through these points and let it meet EΘ extended at K.

ἕως οὗ γένηται · dum … sint

Until sets the stopping condition. Lining up just one contact is insufficient. διήχθω and συμπιπτέτω command a line to be drawn through the points and to meet the extended baseline. Latin's inline page reference is omitted here. In the experiment the straightedge always joins A and Δ, while B and Γ remain independent tests.

Printed pp.108, 109.

5. What the parallel lines establish

ἀλλ᾽ ὡς ἡ ΕΚ πρὸς ΚΖ, ἡ ΑΕ πρὸς ΒΖ, ὡς δὲ ἡ ΖΚ πρὸς ΚΗ, ἡ ΒΖ πρὸς ΓΗ, ὡς δὲ ἡ ΗΚ πρὸς ΚΘ, ἡ ΓΗ πρὸς ΔΘ. καὶ ὡς ἄρα ἡ ΑΕ πρὸς ΒΖ, ἡ ΒΖ πρὸς ΓΗ, καὶ ἡ ΓΗ πρὸς ΔΘ. ηὕρηνται ἄρα τῶν ΑΕ, ΔΘ δύο μέσαι ἥ τε ΒΖ καὶ ἡ ΓΗ.

sed EK : KZ = AE : BZ et ZK : KH = BZ : ΓH, et HK : KΘ = ΓH : ΔΘ. itaque etiam erit AE : BZ = BZ : ΓH = ΓH : ΔΘ. itaque inter AE, ΔΘ duae mediae inuentae sunt, BZ et ΓH.

But as EK is to KZ, so AE is to BZ; as ZK to KH, so BZ to ΓH; and as HK to KΘ, so ΓH to ΔΘ. Therefore AE is to BZ as BZ to ΓH and ΓH to ΔΘ. The two means between AE and ΔΘ have therefore been found: BZ and ΓH.

ὡς … πρὸς … · :

Greek writes the comparisons out; Heiberg's nineteenth-century Latin translation uses ratio symbols and equality signs. The preceding proof first establishes the equal baseline ratios using both vertical and diagonal parallels. The diagram below expands that intermediate step. It is not enough for one pair of triangles to be similar.

Printed pp.108, 109.

6. A physical instrument

ἵνα δὲ καὶ ὀργανικῶς δυνώμεθα τὰς δύο μέσας λαμβάνειν, διαπήγνυται πλινθίον ξύλινον ἢ ἐλεφάντινον ἢ χαλκοῦν ἔχον τρεῖς πινακίσκους ἴσους ὡς λεπτοτάτους, ὧν ὁ μὲν μέσος ἐνήρμοσται, οἱ δὲ δύο ἐπωστοί εἰσιν ἐν χολέδραις,

sed ut per instrumenta etiam duas medias sumere possimus, construatur laterculus ligneus uel eburneus uel aheneus tres tabellulas aequales complectens quam maxime exiles, quarum media adfixa sit, duae autem in strigis moueri possint,

So that we may also take the two means by an instrument, a wooden, ivory or bronze frame is put together, holding three equal plates as thin as possible: the middle one is fitted in place, and the other two can slide in grooves.

πινακίσκους · tabellulas

The diminutive means small panels or plates. ὡς with the superlative λεπτοτάτους means as thin as possible. The passage later asks for parallel, close-fitting motion; Heiberg's p.111 footnote questions how easy this is. The digital model assumes exact rails and zero thickness. Its paper sheet is an exploration aid, not evidence of an ancient surviving instrument or a verified fabrication plan.

Printed pp.110, 111.

7. Bring the plates together

συνάγω δὴ τοὺς ἐν τῷ ὀργάνῳ πίνακας, ἕως ἂν κατ᾽ εὐθεῖαν γένηται τὰ Α, Β, Γ, Δ σημεῖα.

admoueo igitur tabellulas instrumenti, dum puncta A, B, Γ, Δ in eadem linea recta sint.

I bring the plates in the instrument together, until the points A, B, Γ and Δ lie on a straight line.

συνάγω · admoueo

The letter now gives a shorter account said to accompany a dedication. The first-person present describes operating the device; it does not itself establish the accuracy of a physical example. The second figure is on p.112. Heiberg's editorial discussion of the repeated figure remains visible on p.111.

Printed pp.110, 111, 112.

8. One more plate than means

ἐὰν δὲ πλείους μέσας ἐπιταχθῇ εὑρεῖν, ἀεὶ ἑνὶ πλείους πινακίσκους καταστησόμεθα ἐν τῷ ὀργανίῳ τῶν ληφθησομένων μέσων· ἡ δὲ ἀπόδειξις ἡ αὐτή.

sin plures medias sumere iussi erimus, una plures tabellas in instrumento inserentes, quam quot mediae sumendae sunt; et demonstratio eadem est.

If we are instructed to find more means, we shall always place in the instrument one more plate than the means to be taken; the proof is the same.

ἑνὶ πλείους · una plures

ἑνί is a dative of measure: more by one. Three plates give two means and a cube-root ratio. Four would give three means and a fourth-root ratio. This workshop operates the three-plate case only. The Latin's footnote marker is omitted; the remainder of the letter and its poem are available in the scans but are not fully translated here.

Printed pp.112, 113.

Paper, source and model boundaries

The downloadable landscape A4 SVG uses 60 mm square plates and a 25 mm calibration bar. The left loose panel goes above the fixed middle panel; the right goes below it. Print at 100% and verify the bar. The sheet is a modern paper exploration. Physical printing, cutting, assembly and achievable precision have not been checked; it is not a detailed groove fabrication plan.

The model assumes equal panels, exact parallel rails and zero thickness. It has no friction, play, material or structural model. The 0.5 mm displacement in the app is a chosen perturbation, not an ancient accuracy estimate. A mechanical alignment is an operation beyond the straightedge-and-compass tools used in the Euclid course.

J. L. Heiberg, *Archimedis opera omnia cum commentariis Eutocii*, vol. III, Leipzig: Teubner, 1881, pp.102–113. Complete source PDF. This first-edition title page reads MDCCCLXXXI, not 1915. Eutocius transmits the construction within a letter attributed to Eratosthenes. The letter's authorship, dedication and origin stories are not independently established here. The two printed figures appear on pp.109 and 112; pp.110–111 discuss manufacturing and its difficulty.

Selected Greek and Latin excerpts were transcribed against these images. Line wrapping is removed, geometric labels use Greek Unicode and ϑ becomes θ. Inline Latin reference markers are omitted; full scans retain the apparatus and adjacent material. This is not the complete letter. English, word notes, colors, coordinate formulas and the model are modern teaching work. Four plates would give three means and a fourth-root ratio; this model operates the three-plate case.

This HTML embeds all twelve images and its Greek font. It needs no remote subresources and runs no scripts. The links to the application and complete source PDF require a connection.

φέρειαν διὰ τὸ ὀρθὸν ἑστάναι τὸν κύλινδρον. πιπτέτω καὶ ἔστω ἡ KI, καὶ ἡ ἀπὸ τοῦ I ἐπὶ τὸ A ἐπιζευχθεῖσα συμβαλέτω τῇ BZ κατὰ τὸ Θ, ἡ δὲ $A\Delta$ τῷ BMZ ἡμικυκλίῳ κατὰ τὸ M. ἐπεζεύχθωσαν δὲ καὶ αἱ $K\Delta$, MI, $M\Theta$. ἐπεὶ οὖν ἑκάτερον τῶν ΔKA, BMZ ἡμικυκλίων ὀρθόν ἐστι πρὸς τὸ ὑποκείμενον ἐπίπεδον, καὶ ἡ κοινὴ ἄρα αὐτῶν τομὴ ἡ $M\Theta$ πρὸς ὀρθάς ἐστι τῷ τοῦ κύκλου ἐπιπέδῳ· ὥστε καὶ πρὸς τὴν BZ ὀρθή ἐστιν ἡ $M\Theta$. τὸ ἄρα ὑπὸ τῶν ΘB, ΘZ, τουτέστι τὸ ὑπὸ ΘA, ΘI, ἴσον ἐστὶ τῷ ἀπὸ $M\Theta$. ὅμοιον ἄρα ἐστὶ τὸ AMI τρίγωνον ἑκατέρῳ τῶν $MI\Theta$, $MA\Theta$· καὶ ὀρθὴ ἡ ὑπὸ IMA. ἔστιν δὲ καὶ ἡ ὑπὸ ΔKA ὀρθή. παράλληλοι ἄρα εἰσὶν αἱ $K\Delta$, MI, καὶ ἔσται ἀνάλογον, ὡς ἡ ΔA πρὸς AK, τουτέστιν ἡ KA πρὸς AI, οὕτως ἡ IA πρὸς AM διὰ τὴν ὁμοιότητα τῶν τριγώνων. τέσσαρες ἄρα αἱ ΔA, AK, AI, AM ἑξῆς ἀνάλογόν εἰσιν. καί ἐστιν ἡ AM ἴση τῇ Γ, ἐπεὶ καὶ τῇ AB. δύο ἄρα δοθεισῶν τῶν $A\Delta$, Γ δύο μέσαι ἀνάλογον ηὕρηνται αἱ AK, AI.

Ὡς Ἐρατοσθένης.

Βασιλεῖ Πτολεμαίῳ Ἐρατοσθένης χαίρειν.

Τῶν ἀρχαίων τινὰ τραγῳδοποιῶν φασιν εἰσαγαγεῖν τὸν Μίνω τῷ Γλαύκῳ κατασκευάζοντα τάφον. πυθόμενον δέ, ὅτι πανταχοῦ ἑκατόμπεδος εἴη, εἰπεῖν·

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9. ΘB, ΘZ] ΘBZ F; corr. A. 10. ΘA, ΘI] ΘAI F; corr. A; sed fort. scribendum est $B\Theta Z$ et $A\Theta I$. 11. ἄρα ἐστί] utrumque per comp. F. 13. ἔσται] per comp. F. 16. τεσαρες F.

Eutocius / Heiberg 1881 · printed p.103 · Latin · source PDF page 203 (one-based)

cylindrus rectus est.[1]) cadat et sit KI, et linea ab I ad A ducta cum linea BZ in Θ concurrat, et linea $A\Delta$ cum semicirculo BMZ in puncto M. et ducantur etiam lineae $K\Delta$, MI, $M\Theta$. quoniam igitur uterque semicirculus ΔKA, BMZ ad planum propositum perpendicularis est, itaque etiam communis eorum sectio $M\Theta$ ad planum circuli perpendicularis est [Eucl. XI, 19]. quare etiam $M\Theta$ ad BZ perpendicularis est [Eucl. XI def. 3]. itaque

$$B\Theta \times \Theta Z, \text{ h. e. } A\Theta \times \Theta I^2) = M\Theta^2.$$

itaque similis est triangulus AMI utrique triangulo $MI\Theta$, $MA\Theta$ [Eucl. VI, 8 cum $\pi o \rho \acute{\iota} \sigma \mu$.], et angulus IMA rectus. uerum etiam angulus ΔKA rectus est [Eucl. III, 31]. itaque lineae $K\Delta$, MI parallelae sunt, et erit $\Delta A : AK$, h. e. $KA : AI$, $= IA : AM$ propter similitudinem triangulorum.[3]) itaque quattuor lineae ΔA, AK, AI, AM in continua sunt proportione. et $AM = \Gamma$, quoniam $AB = \Gamma$. itaque inter duas datas $A\Delta$, Γ duae mediae proportionales inuentae sunt AK, AI.

Ut Eratosthenes.

Eratosthenes Ptolemaeo regi s. d.

Narrant, quendam tragicum ueterem Minoem Glauco sepulcrum comparantem fecisse; et Minoem, cum comperisset, id undique centum pedes longum

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Hanc epistulam ediderunt Fellius in Arato suo (Oxon. 1672) p. 33 sq., Reimerus de cubi duplicatione p. 146, Bernhardy:

1) Nam K in superficie cylindri est.

2) Eucl. III, 35.

3) $\Delta A : AK = IA : AM$ sequitur ex similitudine triangulorum $AK\Delta$, AMI (cfr. Eucl. VI, 2), $\Delta A : AK = KA : AI$ ex similitudine triangulorum $AK\Delta$, AKI (Eucl. VI, 8).

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μικρὸν γ' ἔλεξας βασιλικοῦ σηκὸν τάφου·
διπλάσιος ἔστω· τοῦ καλοῦ δὲ μὴ σφαλεὶς
δίπλαζ' ἕκαστον κῶλον ἐν τάχει τάφου.

ἐδόκει δὲ διημαρτηκέναι. τῶν γὰρ πλευρῶν διπλασιασθεισῶν τὸ μὲν ἐπίπεδον γίνεται τετραπλάσιον, τὸ δὲ στερεὸν ὀκταπλάσιον. ἐζητεῖτο δὲ καὶ παρὰ τοῖς γεωμέτραις, τίνα ἄν τις τρόπον τὸ δοθὲν στερεὸν διαμένον ἐν τῷ αὐτῷ σχήματι διπλασιάσειεν. καὶ ἐκαλεῖτο τὸ τοιοῦτον πρόβλημα κύβου διπλασιασμός· ὑποθέμενοι γὰρ κύβον ἐζήτουν τοῦτον διπλασιάσαι. πάντων δὲ διαπορούντων ἐπὶ πολὺν χρόνον πρῶτος Ἱπποκράτης ὁ Χῖος ἐπενόησεν, ὅτι, ἐὰν εὑρεθῇ δύο εὐθειῶν γραμμῶν, ὧν ἡ μείζων τῆς ἐλάσσονός ἐστι διπλασία, δύο μέσας ἀνάλογον λαβεῖν ἐν συνεχεῖ ἀναλογίᾳ, διπλασιασθήσεται ὁ κύβος, ὥστε τὸ ἀπόρημα αὐτοῦ εἰς ἕτερον οὐκ ἔλασσον ἀπόρημα κατέστρεφεν. μετὰ χρόνον δὲ τινάς φασιν Δηλίους ἐπιβαλλομένους κατὰ χρησμὸν διπλασιάσαι τινὰ τῶν βωμῶν ἐμπεσεῖν εἰς τὸ αὐτὸ ἀπόρημα. διαπεμψαμένους δὲ τοὺς παρὰ τῷ Πλάτωνι ἐν Ἀκαδημίᾳ γεωμέτρας ἀξιοῦν αὐτοῖς

1. βασιλεικον F; corr. Valckenarius diatribe p. 210. σικον FA. 2. διπλασι cum comp. ον F; corr. Valckenarius. καλοῦ] καλ cum comp. ου F; κύβου B mg. manu 2, ed. Basil., Torellius, al. τοῦ κ. δέ] του δε του κ. F; corr. Valckenarius. σφαλης F; corr. VAD. 3. δίπλαζ'] Nauckius; διπλασιαζ' F; διπλασίαζεν VAD; διπλασιάζων uulgo. 4. δέ] addidi; om. F, uulgo. 5. γίνεται] per comp. F. 6. δὲ καί] scripsi; δη και F, uulgo. 12. Ιπποκρατις F. 16. αυτ cum comp. ου F, uulgo; αὐτό Torellius; αὐτῷ Bernhardy. 17. τιν cum comp. ας F. επιβαλλομενης F; corr. Wurmius cum C; ἐπιβαλομένης νόσου ed. Basil., Torellius, qui post βωμῶν lin. 18 addunt ἐπιταχθέντας. 19. διαμεμψαμενους F, uulgo; corr. F man. 2; διαπεμψομένους ed. Basil., Torellius; διοπομπησομένους uel ἀποδ. Bernhardy. παρά] π (cum comp. αρα) ρα F. 20. αὐτοῖς uulgo.

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esse, dixisse: „paruum quidem indicasti regii sepulcri cubile; duplo maius sit. et pulchrum illud seruans[1]) celeriter singula latera sepulcri conduplica“. uidebatur autem errasse. nam lateribus conduplicatis planum quadruplo maius fit, solidum uero octuplo maius. sed etiam inter geometras quaerebatur, quo modo quis datam figuram solidam in eadem specie manentem conduplicare possit. et hoc problema uocabatur cubi duplicatio; nam cubo proposito hunc conduplicare conabantur. sed omnibus diu haesitantibus primus Hippocrates Chius inuenit, si inter duas lineas rectas, quarum maior duplo maior esset minore, duas proportionales medias in proportione continua sumere didicissemus, cubum conduplicatum iri, ita ut haesitatio eius in haesitationem non minorem conuerteretur. postea autem narrant, Delios quosdam ex oraculo aram aliquam conduplicare conatos in eandem haesitationem incidisse. et legatis missis a geometris in Academia apud Platonem degentibus petisse, ut

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Eratosthenica p. 175 sq., Dressler: E. von der Verdoppelung des Würfels. Wiesbaden 1828, quem librum ex recensione Wurmii (Jahns Jahrbücher XIV p. 185 sq.) sola noui (cum hoc uolumen mihi praesto non esset, inuitus Neue Jahrb. Suppl. XI p. 380 Wurmio et Dresslero coniecturas aliquot eripui, quas hic auctoribus reddidi). fragmentum p. 104 Euripidi (in Polyido) tribuit Valckenarius diatr. p. 210; negat Nauckius Eurip. III p. V, ubi rectissime tertium uersum adiunxit (simile aliquid iam I. N. Loënsis Miscell. epiphyll. III, 1 suspicatus esse uidetur). epigramma Eratosthenis praeterea ediderunt F. Vieta: op. mathemat. (ed. Schooten, Lugd. Batau. 1646) p. 349, Brunckius Anal. I p. 478, Jacobs Anthol. Palat. I^2 p. 315, Reimerus p. 146; Latine uertit P. Ramus: Schol. math. (Francof. 1627) p. 24.

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1) Pulchrum illud, h. e. speciem cubi (*παντὸς τοῦ ἑκατόμπεδος* p. 102, 24).

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εὑρεῖν τὸ ζητούμενον. τῶν δὲ φιλοπόνως ἐπιδιδόντων ἑαυτοὺς καὶ ζητούντων δύο τῶν δοθεισῶν δύο μέσας λαβεῖν Ἀρχύτας μὲν ὁ Ταραντῖνος λέγεται διὰ τῶν ἡμικυλίνδρων εὑρηκέναι, Εὔδοξος δὲ διὰ τῶν καλουμένων καμπύλων γραμμῶν. συμβέβηκε δὲ πᾶσιν αὐτοῖς ἀποδεικτικῶς γεγραφέναι, χειρουργῆσαι δὲ καὶ εἰς χρεῖαν πεσεῖν μὴ δύνασθαι πλὴν ἐπὶ βραχύ τι τοῦ Μενέχμου καὶ ταῦτα δυσχερῶς. ἐπινενόηται δέ τις ὑφ' ἡμῶν ὀργανικὴ λῆψις ῥαδία, δι' ἧς εὑρήσομεν δύο τῶν δοθεισῶν οὐ μόνον δύο μέσας, ἀλλ' ὅσας ἄν τις ἐπιτάξῃ. τούτου δὲ εὑρισκομένου δυνησόμεθα καθόλου τὸ δοθὲν στερεὸν παραλληλογράμμοις περιεχόμενον εἰς κύβον καθιστάναι ἢ ἐξ ἑτέρου εἰς ἕτερον μετασχηματίζειν, καὶ ὅμοιον ποιεῖν καὶ ἐπαύξειν διατηροῦντας τὴν ὁμοιότητα, ὥστε καὶ βωμοὺς καὶ ναούς. δυνησόμεθα δὲ καὶ τὰ τῶν ὑγρῶν μέτρα καὶ ξηρῶν (λέγω δέ, οἷον μετρητὴν ἢ μέδιμνον), εἰς κύβον καθίστασθαι καὶ διὰ τῆς τούτου πλευρᾶς ἀναμετρεῖν τὰ τούτων δεκτικὰ ἀγγεῖα, πόσον χωρεῖ. χρήσιμον δὲ ἔσται τὸ ἐπινόημα καὶ τοῖς βουλομένοις ἐπαύξειν καταπαλτικὰ καὶ λιθοβόλα ὄργανα. δεῖ γὰρ ἀνάλογον ἅπαντα αὐξηθῆναι καὶ τὰ πάχη καὶ τὰ μεγέθη καὶ τὰς κατατρήσεις καὶ τὰς χοινικίδας καὶ τὰ ἐμβαλλόμενα νεῦρα, εἰ μέλλει καὶ ἡ βολὴ ἀνάλογον ἐπαυξηθῆναι· ταῦτα δὲ οὐ δυνατὰ γενέσθαι ἄνευ τῆς τῶν μέσων εὑρέσεως. τὴν δὲ ἀπόδειξιν καὶ τὴν κατασκευὴν τοῦ λεχθέντος ὀργάνου ὑπογέγραφά σοι.

δεδόσθωσαν δύο ἄνισοι εὐθεῖαι, ὧν δεῖ δύο μέσας ἀνάλογον εὑρεῖν ἐν συνεχεῖ ἀναλογίᾳ, αἱ AE, $\Delta\Theta$.

1. ευρ cum comp. ην uel ιν F. 2. ἑαυτούς] εαυτοις F; corr. Dressler; αὐτοῖς Bernhardy. 7. επιβραχυτητι F; corr. A.

sibi inuenirent, quod quaererent. his industrie rem adgressis et inter duas datas lineas duas medias sumere conantibus Archytas Tarentinus dicitur per semicylindros inuenisse, Eudoxus autem per curuas lineas, quae uocantur. accidit autem omnibus, ut demonstrationem perficerent, sed re efficere et usui adcommodare non possent praeter Menechmum paullulum et id operose. a nobis autem methodus per instrumenta habilis inuenta est, qua inter duas lineas datas non modo duas medias sumamus, sed quotcunque quis uoluerit. hoc autem inuento omnino quoduis datum solidum parallelogrammis comprehensum in cubum mutare poterimus aut ex alia in aliam formam conuertere et simile reddere et amplificare similitudinem seruantes, ut etiam aras et aedes. et etiam mensuras humidi et sicci, uelut amphoram et medimnum, in cubum mutare poterimus et per huius latus metiri uasa illas capientia, quantum capiant. et inuentum meum iis quoque utile erit, qui catapultas ballistasque augere uolunt. nam omnia proportionaliter augenda sunt, et crassitudines et magnitudines et foramina et choinicides et funes inserti, si iactus proportionaliter augeri uoluerimus. haec autem fieri nequeunt, quin mediae inueniantur. demonstrationem autem et constructionem illius instrumenti infra tibi exposui.

datae sint duae lineae inaequales, inter quas duas medias proportionales inueniri oportet in proportione continua, AE, $\Delta\Theta$. et in linea $E\Theta$ perpendicularis

8. τις] τηϛ F. 9. ληψ cum comp. ηϛ F. 16. δέ] fort. delendum. 17. ἤ] om. F; corr. Wurmius. μεδιμν cum comp. ων F; corr. Torellius. 21. καί] κι F. 24. μελει F; corr. D. βουλη F; corr. Torellius.

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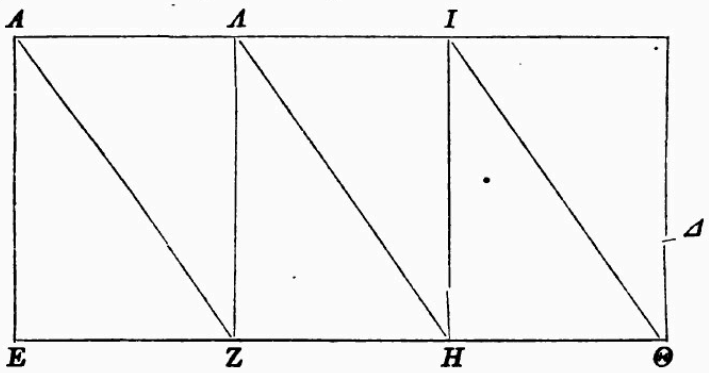
καὶ κείσθω ἐπί τινος εὐθείας τῆς $E\Theta$ πρὸς ὀρθὰς ἡ AE, καὶ ἐπὶ τῆς $E\Theta$ τρία συνεστάτω παραλληλόγραμμα ἐφεξῆς τὰ AZ, ZI, $I\Theta$, καὶ ἤχθωσαν διάμετροι ἐν αὐτοῖς αἱ AZ, ΛH, $I\Theta$. ἔσονται δὴ αὐταὶ παράλληλοι. μένοντος δὴ τοῦ μέσου παραλληλογράμμου τοῦ ZI συνωσθήτω τὸ μὲν AZ ἐπάνω τοῦ μέσου, τὸ δὲ $I\Theta$ ὑποκάτω, καθάπερ ἐπὶ τοῦ δευτέρου σχήματος, ἕως οὗ γένηται τὰ A, B, Γ, Δ κατ' εὐθεῖαν. καὶ διήχθω διὰ τῶν A, B, Γ, Δ σημείων εὐθεῖα καὶ συμπιπτέτω τῇ $E\Theta$ ἐκβληθείσῃ κατὰ τὸ K. ἔσται δή, ὡς ἡ AK πρὸς KB, ἐν μὲν ταῖς AE, ZB παραλλήλοις ἡ EK πρὸς KZ, ἐν δὲ ταῖς AZ, BH παραλλήλοις ἡ ZK πρὸς KH. ὡς ἄρα ἡ AK πρὸς KB, ἡ EK πρὸς KZ, καὶ ἡ KZ πρὸς KH. πάλιν ἐπεί ἐστιν, ὡς ἡ BK πρὸς $K\Gamma$, ἐν μὲν ταῖς BZ, ΓH παραλλήλοις ἡ ZK πρὸς KH, ἐν δὲ ταῖς BH, $\Gamma\Theta$ παραλλήλοις ἡ HK πρὸς $K\Theta$, ὡς ἄρα ἡ BK πρὸς $K\Gamma$, ἡ ZK πρὸς KH, καὶ ἡ HK πρὸς $K\Theta$. ἀλλ' ὡς ἡ ZK πρὸς KH, ἡ EK πρὸς KZ. καὶ ὡς ἄρα ἡ EK πρὸς KZ, ἡ ZK πρὸς KH, καὶ ἡ HK πρὸς $K\Theta$. ἀλλ' ὡς ἡ EK πρὸς KZ, ἡ AE πρὸς BZ, ὡς δὲ ἡ ZK πρὸς KH, ἡ BZ πρὸς ΓH, ὡς δὲ ἡ HK πρὸς $K\Theta$, ἡ ΓH πρὸς $\Delta\Theta$. καὶ ὡς ἄρα ἡ AE πρὸς BZ, ἡ BZ πρὸς ΓH, καὶ ἡ ΓH πρὸς $\Delta\Theta$. εὕρηνται ἄρα τῶν AE, $\Delta\Theta$ δύο μέσαι ἥ τε BZ καὶ ἡ ΓH.

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14. KH] ZH F, Cr. 16. HK] ZK F. 23. ΓH] (alt.) KH F.

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ponatur AE, et in $E\Theta$ tria parallelogramma [aequalia] continua construantur AZ, ZI, $I\Theta$, et in iis diametri ducantur AZ, ΛH, $I\Theta$. hae igitur parallelae erunt. itaque manente parallelogrammo medio ZI inter se



adpropinquent AZ supra medium, $I\Theta$ infra, ut in figura secunda [p. 112], dum puncta A, B, Γ, Δ in eadem linea recta sint. et per A, B, Γ, Δ puncta ducatur linea et concurrat cum $E\Theta$ producta in K. erit igitur

$$AK : KB = EK : KZ$$

in parallelis AE, ZB, et $AK : KB = ZK : KH$ in parallelis AZ, BH. itaque

$$AK : KB = EK : KZ = KZ : KH.$$

rursus quoniam est $BK : K\Gamma = ZK : KH$ in parallelis BZ, ΓH, et $BK : K\Gamma = HK : K\Theta$ in parallelis BH, $\Gamma\Theta$, erit igitur $BK : K\Gamma = ZK : KH = HK : K\Theta$. sed $ZK : KH = EK : KZ$. itaque etiam

$$EK : KZ = ZK : KH = HK : K\Theta.$$

sed $EK : KZ = AE : BZ$ et $ZK : KH = BZ : \Gamma H$, et $HK : K\Theta = \Gamma H : \Delta\Theta$. itaque etiam erit

$$AE : BZ = BZ : \Gamma H = \Gamma H : \Delta\Theta.$$

itaque inter AE, $\Delta\Theta$ duae mediae inuentae sunt, BZ et ΓH.

ταῦτα οὖν ἐπὶ τῶν γεωμετρουμένων ἐπιφανειῶν ἀποδέδεικται. ἵνα δὲ καὶ ὀργανικῶς δυνώμεθα τὰς δύο μέσας λαμβάνειν, διαπήγνυται πλινθίον ξύλινον ἢ ἐλεφάντινον ἢ χαλκοῦν ἔχον τρεῖς πινακίσκους ἴσους ὡς λεπτοτάτους, ὧν ὁ μὲν μέσος ἐνήρμοσται, οἱ δὲ δύο ἐπωστοί εἰσιν ἐν χολέδραις, τοῖς δὲ μεγέθεσιν καὶ ταῖς συμμετρίαις ὡς ἕκαστοι ἑαυτοὺς πείθουσιν· τὰ μὲν γὰρ τῆς ἀποδείξεως ὡσαύτως συντελεῖται· πρὸς δὲ τὸ ἀκριβέστερον λαμβάνεσθαι τὰς γραμμὰς φιλοτεχνητέον, ἵνα ἐν τῷ συνάγεσθαι τοὺς πινακίσκους παράλληλα διαμένῃ πάντα καὶ ἄσχαστα καὶ ὁμαλῶς συναπτόμενα ἀλλήλοις. — ἐν δὲ τῷ ἀναθήματι τὸ μὲν ὀργανικὸν χαλκοῦν ἐστιν καὶ καθήρμοσται ὑπ' αὐτὴν τὴν στεφάνην τῆς στήλης προσμεμολυβδοχοημένον, ὑπ' αὐτοῦ δὲ ἡ ἀπόδειξις συντομώτερον φραζομένη καὶ τὸ σχῆμα, μετ' αὐτὸ δὲ ἐπίγραμμα. ὑπογεγράφθω οὖν σοι καὶ ταῦτα, ἵνα ἔχῃς καὶ ὡς ἐν τῷ ἀναθήματι. τῶν δὲ δύο σχημάτων τὸ δεύτερον γέγραπται ἐν τῇ στήλῃ.

δύο τῶν δοθεισῶν εὐθειῶν δύο μέσας ἀνάλογον εὑρεῖν ἐν συνεχεῖ ἀναλογίᾳ. δεδόσθωσαν αἱ AE, $\Delta\Theta$. συνάγω δὴ τοὺς ἐν τῷ ὀργάνῳ πίνακας, ἕως ἂν κατ' εὐθεῖαν γένηται τὰ A, B, Γ, Δ σημεῖα. νοείσθω δέ, ὡς ἔχει ἐπὶ τοῦ [δευτέρου] σχήματος. ἔστιν ἄρα, ὡς ἡ AK πρὸς KB ἐν μὲν ταῖς AE, BZ παραλλήλοις ἡ EK πρὸς KZ, ἐν δὲ ταῖς AZ, BH, ἡ ZK πρὸς KH.

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3. λαμβαν cum comp. ην uel ιν F. 4. πινακιστους F; corr. BC. 11. ἄσχιστα? 14. προσμεμολυβδοχοημένον] προς μεμολυβδοχημενον F; corr. Dressler (μεμολυβδοχοημένων Bernhardy). 15. συντομοτερον F. 16. μετ' αὐτό] μετὰ τοῦτο Bernhardy. 21. ευρ cum comp. ην uel ιν F. 24. εχη F; corr. Bernhardy. δευτέρου] deleo; u. not. 2.

haec igitur in superficiebus geometricis demonstrata sunt. sed ut per instrumenta etiam duas medias sumere possimus, construatur laterculus ligneus uel eburneus uel aheneus tres tabellulas aequales complectens quam maxime exiles, quarum media adfixa sit, duae autem in strigis moueri possint, ea magnitudine eaque specie, quae cuique placuerit; nam demonstratio eadem ratione conficitur. sed quo subtilius sumantur lineae, opera danda est, ut in admouendis tabellulis omnia parallela maneant et non hiantia et aequabiliter coniuncta inter se.[1]) in dono autem uotiuo instrumentum aheneum est et infra ipsam columnae coronam positum est plumbo adfixum, infra id autem demonstratio est breuius exposita et figura, deinde autem epigramma. itaque haec quoque tibi perscribantur, ut habeas etiam, quae in dono uotiuo sunt. ex duabus autem figuris altera in columna descripta est.[2])

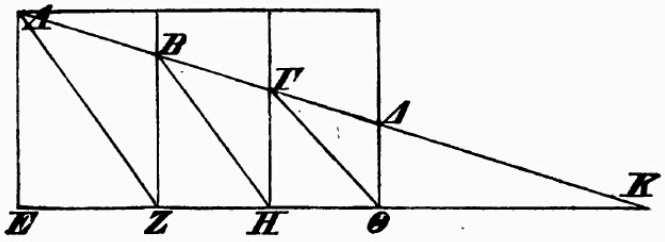
[Ut in dono uotiuo].

Inter duas lineas datas duas medias proportionales inuenire in proportione continua. datae sint AE, $A\Theta$. admoueo igitur tabellulas instrumenti, dum puncta A, B, Γ, Δ in eadem linea recta sint. et fingantur esse, ut in figura sunt. erit igitur $AK : KB = EK : KZ$ in parallelis AE, BZ, et $AK : KB = ZK : KH$ in

1) Sed hoc ipsum difficillimum est; itaque Nicomedes non sine causa instrumentum Eratosthenis uituperauit (infra p. 114); cfr. Sturm p. 110. frustra igitur Eratosthenes gloriatur p. 106, 9 sq.

2) Adparet igitur, in dono unam tantum figuram fuisse (eam, quae hic secunda est (p. 112) et bis ponenda erat); itaque in dono non fuit δευτέρου lin. 24, sed uidetur a librario additum esse.

ὡς ἄρα ἡ *EK* πρὸς *KZ*, ἡ *KZ* πρὸς *KH*. ὡς δὲ αὗται πρὸς ἀλλήλας, ἥ τε *AE* πρὸς *BZ*, καὶ ἡ *BZ* πρὸς *ΓH*. ὡσαύτως δὲ δείξομεν, ὅτι καὶ ὡς ἡ *ZB*



πρὸς *ΓH*, ἡ *ΓH* πρὸς *ΔΘ*. ἀνάλογον ἄρα αἱ *AE*, *BZ*, *ΓH*, *ΔΘ*. ηὕρηνται ἄρα δύο τῶν δοθεισῶν δύο μέσαι. ἐὰν δὲ αἱ δοθεῖσαι μὴ ἴσαι ὦσιν ταῖς *AE*, *ΔΘ*, ποιήσαντες αὐταῖς ἀνάλογον τὰς *AE*, *ΔΘ* τούτων ληψόμεθα τὰς μέσας καὶ ἐπανοίσομεν ἐπ' ἐκείνας, καὶ ἐσόμεθα πεποιηκότες τὸ ἐπιταχθέν, ἐὰν δὲ πλείους μέσας ἐπιταχθῇ εὑρεῖν, εἰ ἑνὶ πλείους πινακίσκους καταστησόμεθα ἐν τῷ ὀργανίῳ τῶν ληφθησομένων μέσων· ἡ δὲ ἀπόδειξις ἡ αὐτή.

Εἰ κύβον ἐξ ὀλίγου διπλήσιον, ὦγαθέ, τεύχειν
φράζεαι καὶ στερεὴν πᾶσαν ἐς ἄλλο φύσιν
εὖ μεταμορφῶσαι, τόδε τοι πάρα, κἂν σύ γε μάνδρην
ἢ σιρὸν ἢ κοίλου φρείατος εὐρὺ κύτος
τῇδ' ἀναμετρήσαιο, μέσας ὅτε τέρμασιν ἄκροις
συνδρομάδας δισσῶν ἐντὸς ἕλῃς κανόνων.
μηδὲ σύ γ' Ἀρχύτεω δυσμήχανα ἔργα κυλίνδρων
μηδὲ Μενεχμείους κωνοτομεῖν τριάδας
δίζηαι, μηδ' εἴ τι θεουδέος Εὐδόξοιο
κάμπυλον ἐν γραμμαῖς εἶδος ἀναγράφεται.

---

10. *εἰ ἑνί*] ἔτι tacite Bernhardy. 11. *μέσων, ταὐτὸν γενήσεται* Torellius. 13. *διπλήσιον*] cfr. Schaeferus ad Gregor. Corinth. p. 527. 14. *φραζεται* FA. *καί*] scripsi; *την* F,

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parallelis AZ, BH. itaque $EK : KZ = KZ : KH$. sed $EK : KZ = AE : BZ = KZ : KH = BZ : \Gamma H$. eodem modo demonstrabimus, esse

$$ZB : \Gamma H = \Gamma H : \Delta\Theta.$$

itaque lineae AE, BZ, ΓH, $\Delta\Theta$ proportionales sunt. itaque inter duas lineas datas duae mediae inuentae sunt. — sin lineae datae lineis AE, $\Delta\Theta$ aequales non sunt, facientes lineas AE, $\Delta\Theta$ cum iis proportionales inter has duas medias sumemus et eas ad illas transferemus [cfr. p. 83 not. 3], et fecerimus, quod iussum erat; sin plures medias sumere iussi erimus, una plures tabellulas in instrumento inserentes[1]), quam quot mediae sumendae sunt; et demonstratio eadem est.[2])

Si cubum paruo molimine conduplicare cogitas, amice, et quamuis speciem solidam in aliud bene conuertere, licet tibi, et saeptum uel sirum uel latam cauernam caui putei hac ratione metiri possis, si intra duas regulas [tabellulas] extremis finibus concurrentes sumpseris medias.[3]) neue confugeris ad Archytae inhabilia molimina cylindrorum neue ad Menechmi lineas illas tres per sectiones conicas efficiendas, neue si qua species linearum curuarum ab Eudoxo diuino con-

1) Sc. *ἐσόμεθα πεποιηκότες τὸ ἐπιταχθέν.*

2) Hoc instrumentum Eratosthenis significat Pappus III, 21 p. 54: *ἐν τῷ Ἐρατοσθένους μεσολάβῳ.* idem III, 23 p. 56—58 summam huius ipsius demonstrationis suis uerbis dat.

3) Ita haec uerba obscura et putida (lin. 18) accipienda puto, maxime propter sequentia: *ἐντὸς δισσῶν κανόνων*; cfr. p. 114 lin. 1. aliae aliorum enarrationes u. apud Bernhardy p. 181, Wurmium p. 191.

uulgo; *ἤ* Iacobs. 15. *τόδε*] *τότε* Wurmius. 16. *σειρον* F; corr. Fellus. 20. *Μεναιχμίους* Proclus in Eucl. p. 111, 23. 22. *εγγραμμαις* F; corr. Brunck, Torellius.

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